

Omar Ibrahim

Junior Penetration Tester | Web & IoT Security

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SUMMARY

Junior penetration tester and IT student with an embedded-systems background and hands-on offensive security practice on TryHackMe (top 8%), Hack The Box and PortSwigger Web Security Academy. Focused on web application testing, enumeration and exploitation, with a growing interest in IoT security. Builds scripts and uses AI tools to speed up recon and reporting, while verifying every finding manually.

SKILLS

Security testing: OWASP Top 10, manual testing, reconnaissance, enumeration, exploitation, networking fundamentals

Tools: Burp Suite, Nmap, Metasploit, SQLmap, Gobuster, OpenVAS, Nessus, Wireshark, IDA

Languages: Python, C, Bash, PowerShell, JavaScript, SQL, x86-64 Assembly

Platforms: Linux, Windows, Git/GitHub, ESP32/ESP8266, Flutter, Firebase, MySQL, Oracle SQL

EXPERIENCE

Penetration Testing & Application Security Trainee – NTI

Aug 2026 – Sep 2026 | Online, Cairo

- Trained in vulnerability assessment, WordPress hardening, penetration testing methodology, cloud security auditing and application security
- Practiced log analysis, threat hunting and security awareness campaign planning
- Completed a real-world capstone project and a freelance security engagement; built a freelance portfolio (Upwork, HackerOne)

Python Developer Intern – CodeAlpha

Oct 2025 – Nov 2025 | Remote, India

- Improved Python automation scripts using OS, Shutil and Regex modules for security tooling and workflow automation
- Built logic-driven applications with advanced data structures and robust error handling

Embedded Software Developer – LABstar

Aug 2023 – Jun 2025 | Cairo, Egypt

- Implemented Role-Based Access Control (RBAC) across multi-user industrial applications, applying least-privilege principles
- Built RFID authentication for time-limited floor access on elevator systems, including secure hardware-to-board communication
- Created a cross-platform IoT controller (Flutter + ESP32 over Wi-Fi) with real-time hardware sync and low-latency commands
- Developed inventory and sales systems on SQL databases with automated audit trails

SECURITY PROJECTS & COMPETITIONS

Penetration Testing Labs – TryHackMe (Top 8%)

Nmap, Metasploit, Gobuster

- Completed rooms on network enumeration, privilege escalation, web application attacks and CTF challenges
- Ranked 4th at the HITU IEEE CTF, competing solo against teams of 3–5

Smart Elevator Interface – Embedded Security

RFID, ESP32

- Designed time-limited floor access with RFID authentication and user authorization logic

OTHER PROJECTS

Bank Management System

Python, CustomTkinter

- Multi-user GUI with RBAC for Customers, Tellers and Managers, plus transaction logging for accountability

Electronics Management System – LABstar

SQL, RBAC

- RBAC-secured multi-user inventory system that blocks unauthorized data access and keeps audit trails

Smart Home with Voice Control (Graduation Project)

Flutter, Firebase, PCB

- Voice-controlled smart home app; led the team on task distribution, PCB design and documentation

Real-Time Sensor Data Pipeline

Python, SQLite, Matplotlib

- Serial data ingestion, SQLite storage and live visualization, applicable to monitoring and anomaly detection

KeyGarden – Password Generator

Python, CustomTkinter

- Generator with auto, text-based, passphrase and batch export modes

Morse Code Converter

x86-64 Assembly

- Win32 API only: O(1) lookup table, manual memory management and direct console I/O (low-level groundwork for reverse engineering)

Cinema Management System

C

- File-based movie, ticket and seat reservation system built with modular programming

EDUCATION

B.Tech, Information Technology – New Cairo Technological University

Expected 2029

High School Diploma, Embedded Systems & AI – MICA, Giza

Jun 2025

- Ranked 3rd nationally in Egypt

LEADERSHIP

- **Technical Project Lead, Graduation Project (MICA, 2025):** led a multidisciplinary team building a voice-controlled smart home with Flutter, Firebase and ESP8266
- **Technical Coordinator, RoboCup 2025 (MICA):** led the Line Follower category, ran technical inspections and mentored teams on sensor calibration

CERTIFICATES

- Cybersecurity Analyst Professional Certificate (14 courses) – IBM, Jan 2026
- Ethical Hacking with OpenVAS & Metasploit – IBM, Jan 2026
- Database Programming with SQL – Oracle, Dec 2024